

Reference History and Target Support in Multi-Window Photovoltaic Forecasting

Jiangkun Zhu,^{1, a)} Mengling Yang,¹ Zhicong Chen,^{1, 2, b)} and Lijun Wu^{1, 2, c)}

¹⁾ College of Advanced Manufacturing, Fuzhou University, 1 Shuicheng Road, Quanzhou 362251, China

²⁾ College of Physics and Information Engineering, Fuzhou University, 2 Xueyuan Road, Fuzhou 350116, China

Forecasting gains depend on the reference history, the scored power range, and the evaluation period. We examine these dependencies in five photovoltaic systems at three geographic sites, using 60 frozen neural runs and two linear information sets: six-hour recent history, with or without an available previous-day trajectory. Forecasts are assessed over cumulative one- to twelve-hour windows. A post hoc expanded regularization grid brings Yulara daily-augmented Ridge to 15.84 kW twelve-hour error, close to Daily at 16.07 kW and Inverted-variate at 15.81 kW; paired temporal intervals cross zero for both contrasts. Fixed next-year evaluation at the two external sites changes these comparisons without refitting. At Yulara, original-grid Ridge, Daily, expanded-grid Ridge and Inverted-variate have full-target errors of 8.02, 9.56, 10.84 and 13.86 kW, respectively. At NIST, Inverted-variate improves on Daily in that period, reversing the earlier full-target ordering. Separately, complete-window conditioning in Qcells strongly changes target-power composition; the finite-negative exclusion rule lacks confirmed provider justification. These results show why information gains, target selection and time-period effects must be distinguished. They support conditional comparisons of frozen forecasters, rather than a universally preferred model or regularization grid.

Keywords: photovoltaic forecasting, reference information, cumulative forecast windows, error decomposition

I. INTRODUCTION

Photovoltaic (PV) forecasting supports storage scheduling and power-system operation by estimating how generation will change after a forecast is issued. The practical question is what a learned forecast adds to information already available at that time. Repeating the latest power and retrieving yesterday's corresponding power supply different references. A gain over the former need not imply a gain over the latter. Moreover, aggregate error combines periods of substantial production with low-output targets, whose frequencies and errors can change the comparison.¹⁻³

We study these dependencies through a co-located comparison at Alice Springs and two independently fitted external-site replications. Five target systems cover three geographic sites, with separate models and pre-processing for each system. Four compact neural implementations use six hours of observations to predict a twelve-hour trajectory. The original neural experiment is supplemented by prediction-only analyses and two deterministic linear references: Ridge using the same recent history, and Ridge augmented with an available previous-day trajectory. These additions are post hoc analyses of performance splits already observed during the project.

The principal finding concerns the interpretation of forecasting gains after accounting for available history and scored target support. Daily Persistence changes

the conclusion drawn from Last-value Persistence at Alice Springs; the external primary model usually improves on both, but its NIST twelve-hour full-target comparison reverses despite lower power-active error. An additive squared-error decomposition quantifies that reversal. Common-origin and temporal-block analyses then separate differences in target support and training randomness from variation over the observed dates. The linear references isolate the value of added daily history within one model family. Their missingness and regularization diagnostics distinguish an unstable reference design from a failure of simple learning itself. This distinction, together with explicit support attribution, is central to interpreting the reported gains.

II. RELATED WORK

PV forecasting research uses historical measurements, physical models, numerical weather prediction, sky imagery, and their combinations. These information sets support different forecasting tasks.^{1,2,4} Weather-based benchmarks compare classical and deep models under future weather inputs; geographically distributed studies also investigate transfer.^{5,6} Their numerical errors are not interchangeable with those of a history-only, separately fitted experiment. Studies using Alice Springs and Yulara establish the value of these archives, while differing in periods, available fields, and reference definitions.⁷⁻⁹

Recurrent, variate-token, patch-token, and convolutional models provide alternative ways to summarize a measured history.¹⁰⁻¹³ Here these mechanisms are represented by compact project implementations with a fixed training budget. Their comparison addresses the imple-

^{a)}ORCID: 0009-0009-5335-2345

^{b)}Electronic mail: zhicong.chen@fzu.edu.cn

^{c)}Electronic mail: lijun.wu@fzu.edu.cn

mented forecasters, rather than optimally tuned official architectures. Forecast-evaluation guidance motivates chronological splits, Train-only transformations, and explicit sample support.^{14–17} We use these controls as the basis for interpreting reference-dependent error, rather than as an algorithmic contribution.

III. STUDY DESIGN AND INFORMATION SETS

A. Five targets at three sites

Figure 1 shows the information available at forecast time. Alice Springs contributes three co-located arrays and 36 historical neural runs; Yulara and NIST Ground each contribute one target and 12 runs. Seeds are 42, 43, and 44. No observations are pooled across systems. Table I records the two input regimes and chronological splits.

The Alice targets are Sanyo hybrid-silicon, Hanwha polycrystalline-silicon, and Qcells monocrystalline-silicon arrays.^{18,19} Eight numerical variables comprise array power and Performance Ratio, ambient temperature, relative humidity, and global/diffuse horizontal/tilted radiation. Eight original missingness masks and one Isolation Forest flag bring the input width to 17. Co-location provides shared weather context; differences between arrays also include their electrical systems, orientation, and sensing. Technology is therefore not an isolated experimental treatment.

The Yulara target is a 106.6 kW combined facility, not multiple independent sites.²⁰ Both external systems use seven channels in a fixed order: historical AC power, ambient temperature, global horizontal irradiance (GHI), the three corresponding missingness masks, and an Isolation Forest flag. This common available information set is fitted independently at each site. It does not transfer the original 17-channel weights. Yulara’s temperature is used as the provider’s Celsius field and GHI in its native numerical units, with no additional conversion or cross-site absolute irradiance inference.

NIST power comes from the AC real-power meter; its inverter-power field, including the -999 code, is excluded. Temperature and GHI use the official averaged fields, with the supplied W/m^2 field used directly.^{21–23} The fixed EST/LST coordinate retains UTC $-05:00$ throughout, without daylight-saving changes. At each five-minute endpoint T , only observations in $[T - 5 \text{ min}, T)$ contribute. A variable requires five distinct finite minute values; otherwise its aggregate is missing. Yulara retains the provider-local coordinate with a conservative five-minute availability delay. Two off-grid records are excluded without rounding. These availability conventions prevent a completed measurement interval from entering an earlier input.

Finite negative power and GHI, including unusual operating conditions, remain in the external records. The original Alice rule retains finite nonnegative labels. Miss-

ing labels are never filled. Neither low output nor a power–GHI discrepancy is used to remove Test records. The Supplement gives field definitions, time audits, and the unavailable NIST operation log and partial Yulara metadata.

B. Preprocessing and forecast definition

Original numerical missingness is recorded before input filling. A five-neighbor KNN imputer is fitted on Train. Isolation Forest then uses the imputed Train values (100 trees, contamination 0.01, random state 42); its flag and the original masks are appended. A feature MinMaxScaler and a separate target scaler are fitted on Train only, the latter using valid unimputed power.^{24,25} Validation and Test use the fitted transformations unchanged. Missing timeline positions remain in place.

At origin t , the 72 input positions end at t , and their values must be available by t . The first target is $t + 5$ min and the final target $t + 12$ h. Inputs and labels remain within one split. H12, H48, H96, and H144 denote *cumulative forecast windows*: the first 12, 48, 96, and 144 targets, corresponding to 1, 4, 8, and 12 h. They are not errors at four isolated leads. Original Test evaluation retains origins with valid labels throughout the requested prefix, so different windows can have different origins. A separate common-origin analysis uses complete H144 origins for every prefix; fixed-lead curves use these same origins and score one lead at a time.

IV. FORECASTERS AND MATCHED EVALUATION

A. Compact neural implementations

The recurrent forecaster combines multi-dilation convolutions, an attention-encoder branch, gated fusion, and an attention-weighted bidirectional GRU summary. Its trajectory decoder obeys

$$s_k = \text{GRUCell}(c, s_{k-1}), \quad \hat{y}_k = \text{readout}(s_k), \quad s_0 = c. \quad (1)$$

The fixed context c is supplied at every step. Previous predicted power is not fed back, so this is recurrent hidden-state trajectory decoding, rather than power-autoregressive forecasting. The other three implementations use direct multi-output heads. Inverted-variate embeds each channel’s complete history as a token; Joint-patch embeds overlapping multivariate patches; Depthwise TCN uses an input projection and depthwise/pointwise temporal blocks. Layer specifications and parameter counts for both input widths are in the Supplement. All reported rankings refer to these compact implementations.

Historical neural training used batch size 256, AdamW with learning rate 10^{-3} and weight decay 10^{-5} , gradient clipping 1.0, a maximum of 25 epochs, and early-stopping

Alice: 3 arrays / 17 channels Yulara and NIST: 1 system each / 7 channels

Independent system fits; no pooling or zero-shot transfer

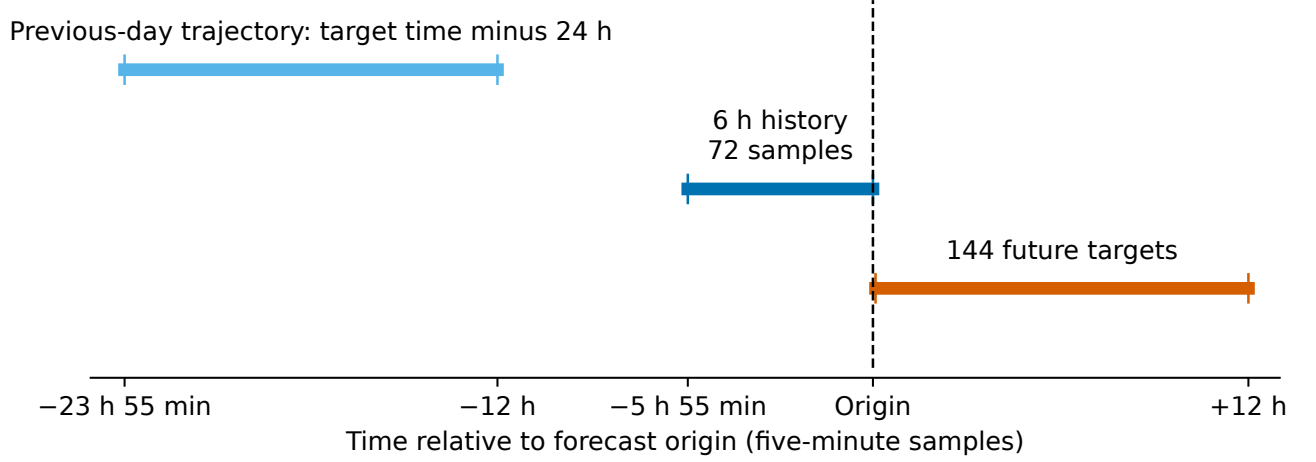


FIG. 1. Every future target at origin plus j times five minutes uses its exact minus-24-hour historical counterpart for Daily or Ridge B. The six-hour history contains 72 samples from minus 355 minutes through origin; the twelve-hour target sequence begins at plus five minutes.

TABLE I. Three geographic sites and two input regimes. Capacities are descriptive DC ratings; error normalization uses each target’s Train range. All forecast grids are five-minute grids. Dates in the last two rows refer to 2017, and those in the first row to 2018.

Site / systems	DC rating / channels	Train; Validation; Test	Source resolution / time basis	Interpretation limit
Alice Springs / three co-located arrays	Sanyo 6.30, Hanwha 5.83, Qcells 5.90 kW 17 channels	Apr 1–Jul 15; Jul 16–Aug 7; Aug 8–31	5 min / frozen ACST coordinate	Shared weather; technology is not an isolated treatment
Yulara / one combined facility	106.6 kW / 7	Jan–Aug; Sep; Oct–Dec	5 min / provider local; +5 min availability	GHI metadata and UTC offset partly unconfirmed
NIST Ground / one array	271 kW / 7	Jan–Aug; Sep; Oct–Dec	1 min to 5 min / fixed EST, UTC–05:00, no DST	Complete 2017 operation log unavailable

patience five with minimum improvement 10^{-8} . Training and checkpoint selection require complete H144 targets. The selected checkpoint minimizes Validation total squared error divided by total valid-target count, not an unweighted average of batch losses. This common budget is a controlled comparison, not evidence that every architecture is fully optimized. In particular, all three NIST TCN runs have their best Validation checkpoint at epoch 25; the historical learning curves reach the imposed boundary.

B. References and target-power scopes

Last-value repeats the true valid power at the origin. Daily Persistence retrieves power at each target timestamp minus exactly 24 h through a timestamp join. At the longest lead, this observation is still twelve hours before the origin. Daily thus uses a different history from the six-hour neural input. On Daily-matched comparisons, every method uses the same origins, labels, and elementwise intersection of valid target and Daily-lag points. This ensures identical scored targets, not identical information.

The full scope includes all eligible targets. The *power-active* scope uses true future power greater than 1% of that system’s maximum Train power. This is a descrip-

tive, post hoc evaluation mask, not a forecast-time detector. Its complement is *low-power*; it may include low production during daytime. The mask does not identify night, snow, maintenance, or outages. Frozen machine-readable records retain the historical key `daylight`; its numerical definition and results are unchanged.

C. Metrics and analysis hierarchy

RMSE, MAE, bias, and R^2 use the stated common targets. Train-range nRMSE divides RMSE by that system's Train power range. Reported nRMSE and skill use percentages, with skill $100(1 - \text{RMSE}_m/\text{RMSE}_r)$; a zero reference error leaves skill undefined. Negative R^2 is retained, and zero target variance leaves it undefined. Neural summaries average per-seed metrics, with sample SD (ddof = 1), rather than evaluating a mean-prediction ensemble. Baselines are deterministic. Seed SD describes training randomness, not temporal sampling uncertainty.

Inverted-variate was fixed as the external primary model from the frozen Alice comparison before external Test predictions were generated. It is a descriptive focal model at Alice. Four-model rankings are secondary. The post hoc minimum mean neural RMSE is a descriptive envelope favorable to the neural group, not a prespecified or deployable forecast. Win counts across windows and scopes are dependent descriptive summaries. We do not pool cross-site kW errors or add the Alice 24 and external 16 combinations into an overall win total.

D. Post hoc explanatory analyses and linear references

The additional analysis plan was recorded before calculating its new scores, after the original Test results were known. The power decomposition uses disjoint active and low-power targets:

$$\text{MSE}_{\text{full}} = \frac{N_a}{N} \text{MSE}_a + \frac{N_l}{N} \text{MSE}_l. \quad (2)$$

Contributions are computed per seed and then averaged. Common-origin sensitivity and fixed-lead errors distinguish window length from changing sample composition. Monthly external summaries assign complete trajectories to the forecast-origin month. Calendar cases use the first legal origin on or after the 15th at 10:00 in each Test month; additional best/worst cases are explicitly post hoc diagnostics, always using seed 42.

For temporal uncertainty, paired nonoverlapping 48 h origin blocks are sampled with replacement 2000 times, with 24/72 h sensitivity checks and fixed random seed 20260908. The same block indices are used for methods and neural seeds. Each resample sums squared errors and counts before computing per-seed RMSE and skill, then averages the seed skills. Percentile 95% intervals are conditional on observed dates and fitted models. Blocks preserve local dependence but cannot remove

all dependence between overlapping trajectories across block boundaries; the short Alice Test period provides especially limited temporal support.

Ridge A flattens the corresponding 72-step, 17- or seven-channel processed neural history. Ridge B adds 144 exact previous-day target-time power values and their missingness masks. Missing lag inputs use Train-column medians. Standardization is Train-fitted, labels are not imputed, and both variants use complete H144 Train/Validation windows. One common regularization parameter from $\{0.1, 1, 10, 100, 1000\}$ minimizes full H144 Validation squared error, with larger regularization breaking an exact tie. The solver is deterministic and uses no seed repetitions. Test uses the original horizon-specific origins and common Daily-valid masks. These references test recent-history learning and additional daily information; they do not predict the effect of giving neural models the same added history.

The later diagnostic revision preserves these original fits. It reconstructs their actual standardized matrices and compares the spectral solution with an independent Cholesky solution of $\|Y - XB\|_F^2 + \alpha\|B\|_F^2$, retaining the same unpenalized intercept and scaled targets. After seven of ten selections were found at original grid endpoints, a uniform $10^{-4}, 10^{-3}, \dots, 10^8$ grid was fixed for all systems and both variants. Its selections use the same full-H144 Validation criterion; no clipping, Test-driven threshold, or new neural fit is introduced. Expanded-grid results remain a separate sensitivity analysis.

Ridge temporal comparisons use a common complete-H144 origin set and the Daily-finite point intersection for A, B, Daily and all three Inverted seeds. The effect is $\text{RMSE}_B - \text{RMSE}_{\text{reference}}$, so negative values favor B. Each neural-seed contrast is computed first, and their mean uses the same temporal draw. These intervals quantify time-sample variation in frozen predictions, not uncertainty from retraining. Qcells support attribution separately records overlapping exclusion flags and an exclusive order: history/tail boundary, future missing or nonfinite labels, future negative labels, invalid origin power, then retention. Daily missingness remains a separate point intersection.

E. Frozen next-year evaluation and scoring sensitivity

Before calculating the new-period errors, forecast origins were fixed to 1 April–30 June 2018 at Yulara and NIST. The final origin is 30 June at 23:55 and its last target is 1 July at 11:55. Source data from 30 March through 1 July provide historical and target buffers. NIST retains fixed EST and strict five-minute aggregation; Yulara retains provider-local availability coordinates. Original 2017 processors, three Inverted checkpoints and both saved Ridge grids are applied without fitting any component or changing thresholds. This is a same-site next-year fixed-period evaluation, not zero-shot transfer to new sites or year-round validation.

Each new-period comparison scores the intersection of finite target power, valid origin power and available Daily values. Missing labels are never filled; negative finite external power remains eligible. This pointwise support is stated separately from the original complete-H144 Test support. Differences between periods therefore are not attributed solely to season. The same 48-hour primary and 24/72-hour sensitivity block procedure applies, with origin blocks anchored at 1 April.

For Alice, a separate support analysis distinguishes pointwise nonnegative targets (S1) from finite raw targets (S2), retaining the frozen input and Daily rules. Provider documentation does not establish that every finite negative measurement is invalid. A completed-window filter and a pointwise scoring mask answer different questions. The restored-origin neural comparison is not promoted here because the new forward path has not reproduced every historical checkpoint within the original numerical tolerance; its support counts and diagnostic status are reported in the Supplement.

V. RESULTS

A. Temporal support and simple learned references

Paired 48 h block intervals qualify the small long-window Daily gains. Yulara twelve-hour/full skill is 1.62%, with a conditional interval of $[-9.47, 9.85]\%$; NIST eight-hour/full is 3.88% with $[-6.10, 11.91]\%$, and twelve-hour/full is -6.47% with $[-17.14, 2.28]\%$. NIST's corresponding power-active eight- and twelve-hour intervals are $[11.75, 27.07]\%$ and $[4.60, 19.94]\%$. All prechosen block-size results are retained in the Supplement. These intervals describe temporal variation over the observed Test support and fitted models, separately from seed SD.

1. Saved-grid effects and paired intervals

Table II presents both saved grids on identical twelve-hour targets. Ridge B reduces full-target error relative to A in all five systems, but this is an information-set comparison within a particular linear specification. Figure 2 now shows the expanded-grid intervals; original-grid intervals are retained in the supplementary evidence. At NIST, B minus Daily is -6.90 kW with a 48-hour interval $[-8.43, -5.31]$ kW; B minus mean Inverted is -9.61 kW with $[-12.08, -7.15]$ kW. At Qcells, the observed -0.0071 kW B-minus-Daily difference has interval $[-0.0525, 0.0494]$ kW. These are conditional differences over the scored dates, not proof of stability across future years. All 24/72-hour checks, power-active comparisons and individual neural-seed contrasts are retained.

TABLE II. Saved original and expanded-grid Ridge twelve-hour/full RMSE (kW), on identical complete-H144 Daily-matched targets. Both grids are post hoc.

System	Ridge A		Ridge B	
	Original	Expanded	Original	Expanded
Sanyo	0.914	0.914	0.212	0.212
Hanwha	0.844	0.844	0.248	0.248
Qcells	0.839	0.839	0.237	0.259
Yulara	119.585	29.838	35.950	15.842
NIST	48.956	48.956	34.995	34.995

2. Why the Yulara reference extrapolates

Yulara Ridge A's original Validation MSE is 47033.79 kW² (216.87 kW RMSE), while its Train maximum target is 109.90 kW. An unconstrained linear predictor can exceed this range. Reconstruction reproduces the saved Test predictions, and the actual-matrix Cholesky comparison differs by less than 9×10^{-9} kW. Alignment, units and inverse scaling do not account for the large error.

The decisive diagnostic concerns missingness exposure. At the origin position, temperature and GHI missing indicators occur in four of 69333 eligible Train histories (0.0058%), versus 18.56% in Validation and 5.50% in Test. Their Train standard deviation is 0.007595, so an active mask becomes about 132 standardized units. The two identical weather-mask channels each contribute about 107.64 kW RMS to the Validation prediction and 58.43 kW RMS to Test. These are additive prediction components, not independent error contributions. Exposure to long missing patterns combines with coefficients estimated from very sparse Train occurrences. The largest-error 5% of Test origins account for 90.68% of A's SSE over available saved targets; no such origins are deleted. The comparable main-table score instead uses complete-H144 Daily-matched support.

The expanded grid reduces Yulara's Daily-matched full-target errors from 119.58 to 29.84 kW for A and from 35.95 to 15.84 kW for B, with Validation selecting 10^7 and 10^6 . This supports a missingness-extrapolation and regularization-range explanation, rather than a solver failure or a general inability of linear models to use daily information. All ten original and expanded fits are shown in the Supplement, including cases where a lower Validation error does not lower Test error. No expanded-grid result replaces the original-grid temporal intervals or the frozen neural conclusions.

B. Complete-window conditioning selects different power targets

Qcells has 6463 one-hour Test origins and 77556 target pairs, including 36504 power-active pairs (47.07%). Requiring complete twelve-hour labels leaves 2996 origins and 35952 one-hour pairs, of which 42 are active

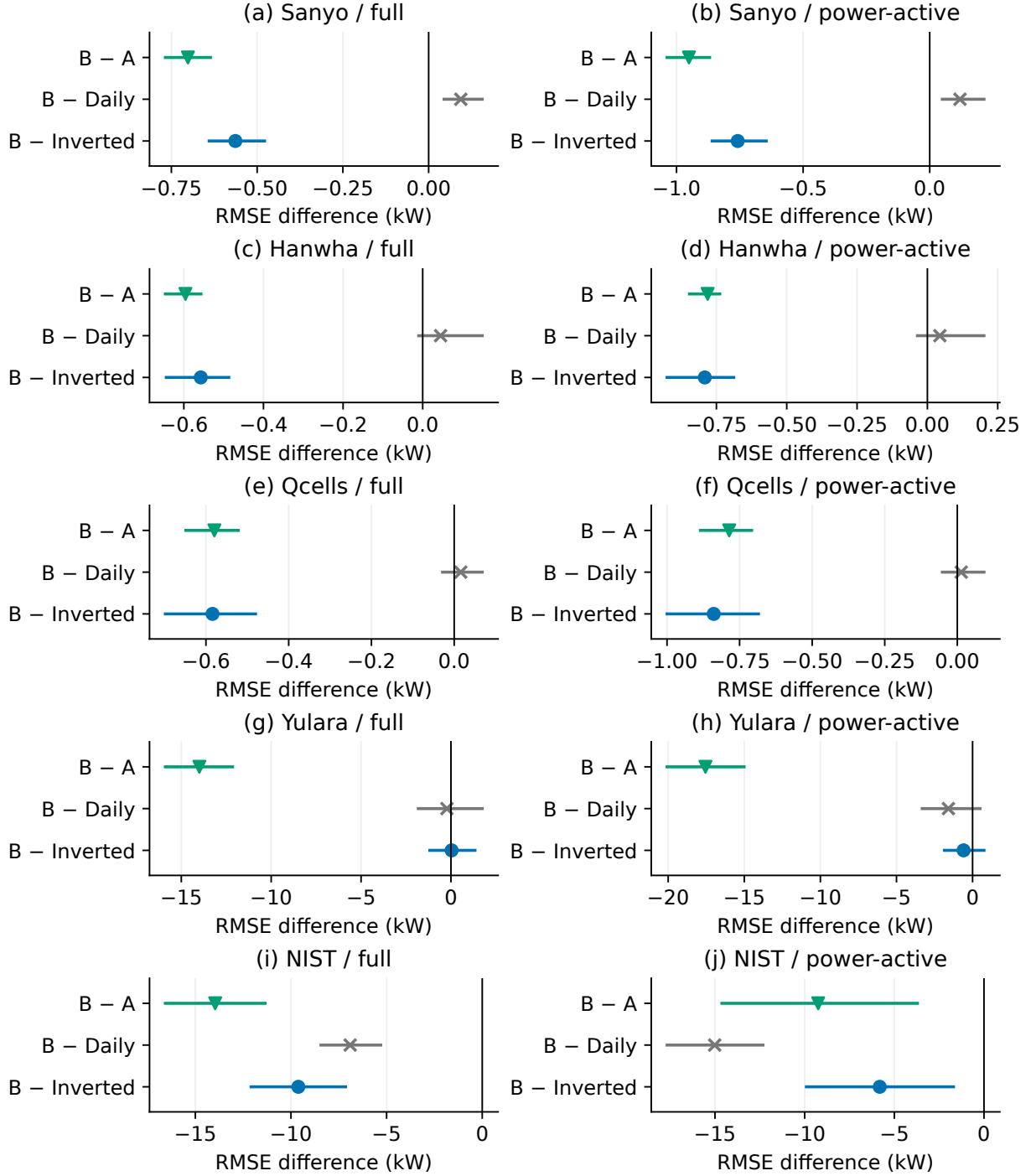


FIG. 2. Expanded-grid Ridge B minus each reference over cumulative 12 h windows. Negative differences favor B. Bars are 95% paired 48 h origin-block intervals, 2000 draws; Inverted contrasts average three seed-specific effects, not predictions. Identical complete-H144 and Daily-finite target support is used within each system/scope. Scales differ between panels. Original-grid intervals remain in the supplementary evidence.

(0.117%). Those 42 pairs involve 25 distinct physical target timestamps and 25 origins; they are not 42 independent observations. The complete subset contains 3260 distinct one-hour target timestamps overall, versus 6760 under the original one-hour support.

The exclusion mechanism is identifiable (Fig. 3). Among the 3467 removed one-hour origins, the exclusive accounting assigns 117 to the split tail and 3350 to a finite negative value later in the twelve-hour target window; no future nonfinite labels contribute. The overlapping negative-value flag marks 3415 origins, including 65 already assigned to the tail. The raw Test series has 54 finite negative records, 52 at local hour 18 and two at hour 7. Their physical cause is not inferred. The frozen nonnegative-label rule spreads these sparse records across many overlapping long windows. Removed origins have an 87.64% active share in their first hour, explaining why the retained subset is dominated by low-power targets.

The mechanism also operates before Test. On complete-H144 evaluation support, the active share of first-hour pairs is 3.08% in Train and zero in Validation, versus 44.84% and 44.96% on one-hour support. Historical Alice neural training checks full future-label validity without requiring valid origin power, unlike evaluation and Ridge fitting; its corresponding Train and Validation counts are 12747 and 2999 rather than 12648 and 2977. This distinction is reported without changing frozen models. Common-origin and fixed-lead curves therefore describe a selected complete-window population. Last-value’s 0.00398 kW versus Inverted’s 0.12494 kW on the retained one-hour full targets does not establish performance over the removed production periods.

C. Explaining the NIST full-target reversal

NIST twelve-hour Daily-matched support contains 3128676 target points, of which 1098178 are power-active and 2030498 (64.90%) are low-power. Inverted-variate’s mean per-seed RMSE is 44.61 kW overall, 60.54 kW active, and 32.91 kW low-power. Daily’s corresponding errors are 41.90, 69.75, and 8.60 kW. The active subset favors the neural forecast, while the larger low-power subset carries a substantial extra error.

Table III and Fig. 4 quantify this cancellation on the squared-error scale. The neural model reduces the active contribution to full MSE by 420.64 kW² but increases the low-power contribution by 655.38 kW², leaving an overall increase of 234.74 kW². The neural low-power bias is positive (12.73 kW), compared with 1.11 kW for Daily. This directly explains the aggregate reversal through scored target subsets; it does not identify a weather event or a network mechanism. The decomposition is also supplied for every system, model, and window.

TABLE III. NIST twelve-hour contributions to full MSE (kW²). Contributions, not RMSEs, add across disjoint target subsets. Neural values average per-seed contributions.

Method	Active	Low-power	Total
Inverted	1286.91	703.40	1990.31
Daily	1707.55	48.02	1755.56

D. Alice Springs: the reference changes the comparison

The frozen primary winners remain Inverted-variate 12, TCN 9, Joint-patch 2, recurrent 1, and Last-value 0 across 24 array-window-scope combinations. Table IV retains the short and full-window primary summary. The neural envelope improves on Last-value throughout, but Daily has lower matched error than that favorable envelope in 22 of 24 combinations. The two neural exceptions are Hanwha’s one-hour full and power-active comparisons. Figure 5 places both references on the same Daily-valid targets, so their contrast cannot be explained by using different target subsets for the two baselines.

E. External replications: recurrent gains, conditional rankings

The external prespecified model has lower mean RMSE than Last-value in all 16 comparisons and than Daily in 15. Table V and Fig. 6 retain every comparison; the table gives seed dispersion and the figure gives conditional temporal intervals. At twelve hours/full, Last-value skill is 67.10% at Yulara and 43.24% at NIST; Daily-matched skill is 1.62% and −6.47%, respectively. The reference therefore changes both the size and, at NIST, the sign of the observed advantage.

Inverted-variate leads all eight Yulara neural comparisons. At NIST its mean neural rank is 2.25: recurrent decoding leads the full scope, while TCN leads the longer power-active windows. The Supplement shows neural error gaps from the best mean without amplifying small differences through ranks alone. The post hoc neural envelope beats Daily in eight Yulara and seven NIST comparisons, preserving NIST twelve-hour/full as the Daily exception. These rankings describe the fixed budget and do not establish a site-invariant ordering of architecture families.

F. Same-site next-year fixed-period results

Figure 7 compares the frozen models with Daily in the fixed April–June 2018 origin period. At Yulara, twelve-hour/full RMSE is 8.02 kW for original-grid B, 9.56 kW for Daily, 10.84 kW for expanded-grid B, and 13.86 kW for mean Inverted. Original B minus Daily is −1.54 kW with a 48-hour interval [−1.92, −1.13] kW; expanded B minus Daily is +1.28 kW with [0.42, 2.27] kW. Inverted

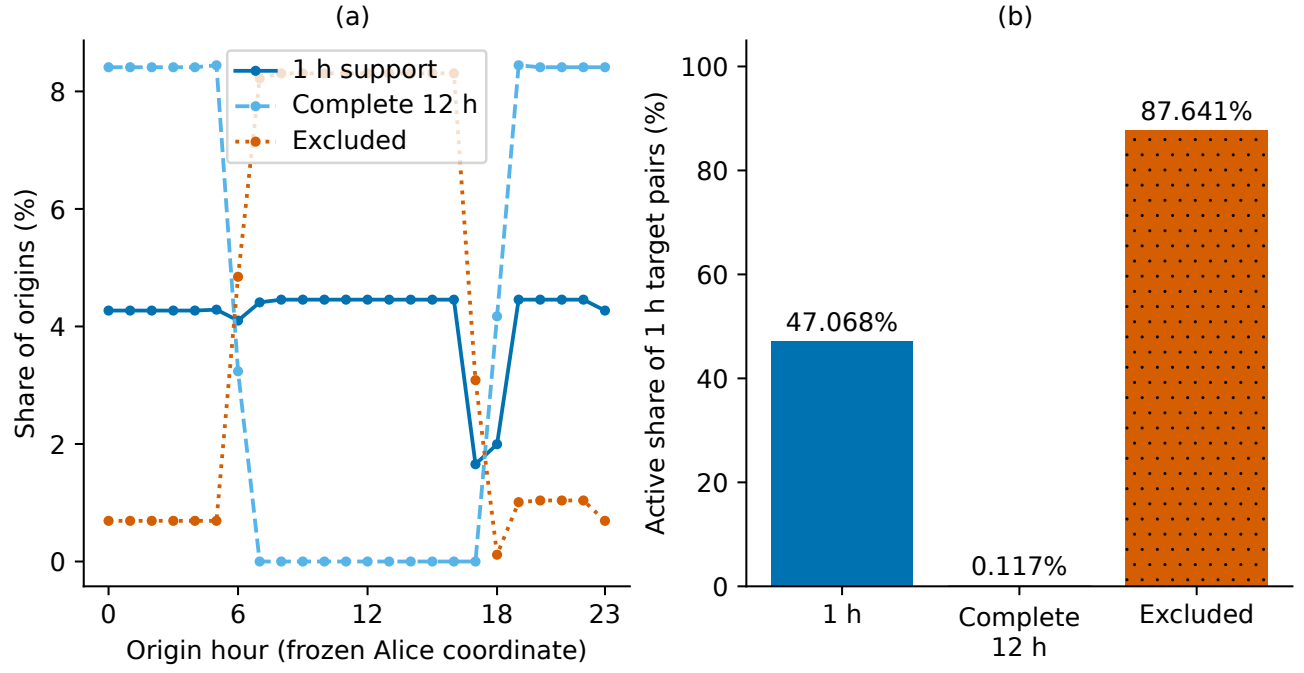


FIG. 3. Qcells Test: 6463 one-hour origins become 2996 complete-twelve-hour origins. Exclusive attribution assigns 3350 removals to future negative labels and 117 to the tail boundary. Active one-hour pairs decrease from 36504 to 42, involving 25 unique active timestamps and 25 origins.

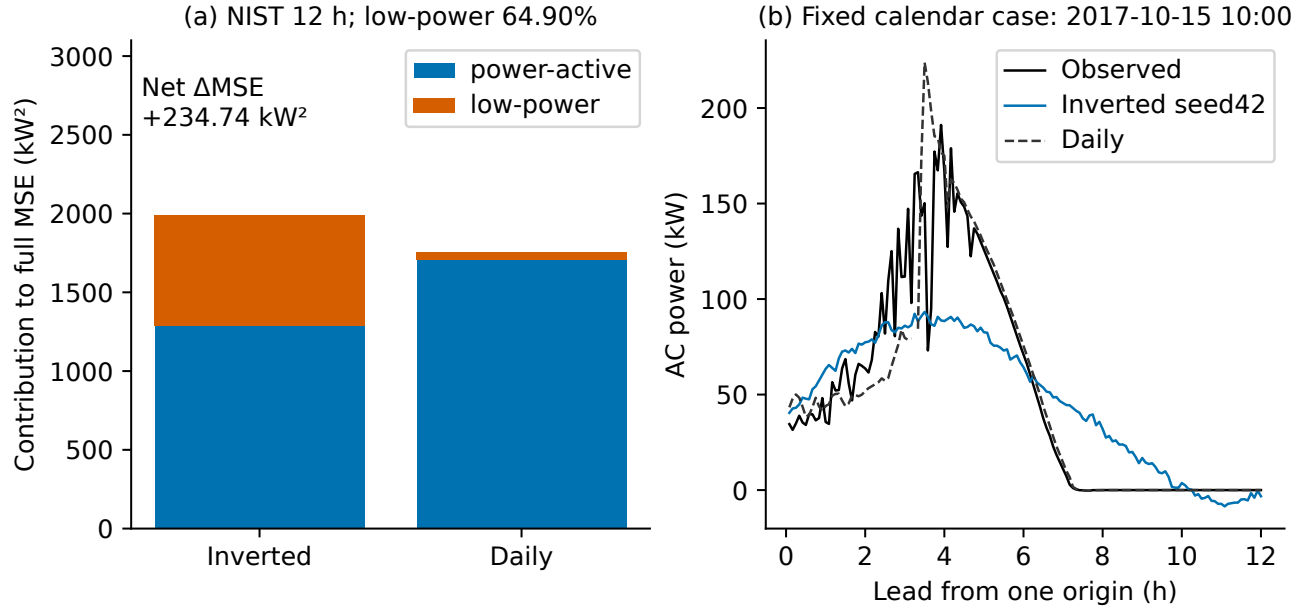


FIG. 4. Left: additive active/low-power contributions to full MSE on NIST twelve-hour Daily-matched targets; neural contributions are averaged over seeds. Active and low-power Inverted-minus-Daily MSE contributions are -420.64 and +655.38 kW²; net +234.74 kW². Right: seed42 at 2017-10-15 10:00 fixed EST, chosen by the calendar rule, not a rolling reconstruction.

minus Daily is +4.30 kW with [2.93, 5.65] kW. Thus neither the earlier neural advantage over Daily nor the expanded-grid improvement over original B persists in this period. The two saved grids remain diagnostic com-

parisons, without reselection from these errors.

At NIST, mean Inverted has 38.19 kW RMSE, both B grids have 40.78 kW, and Daily has 51.03 kW. Inverted minus Daily is -12.84 kW with [-17.40, -8.00] kW. B

TABLE IV. Primary Test Train-range nRMSE and skill (%) on horizon-specific valid origins. The envelope is the post hoc minimum three-seed mean among the four compact implementations and is a descriptive upper bound, not a prespecified model; the complete model-by-seed table is in the Supplementary Material.

Array	Scope	Window	Envelope member	Envelope nRMSE	Last-value nRMSE	RMSE skill
Sanyo	Full	1 h	Discrete recurrent	3.42	7.66	55.30
Sanyo	Full	12 h	Inverted-variate	12.80	45.06	71.60
Sanyo	Power-active	1 h	Depthwise TCN	4.80	11.01	56.40
Sanyo	Power-active	12 h	Inverted-variate	17.45	60.56	71.20
Hanwha	Full	1 h	Depthwise TCN	3.01	7.45	59.50
Hanwha	Full	12 h	Depthwise TCN	12.75	43.94	71.00
Hanwha	Power-active	1 h	Depthwise TCN	3.96	10.78	63.30
Hanwha	Power-active	12 h	Depthwise TCN	17.82	56.14	68.30
Qcells	Full	1 h	Inverted-variate	5.42	7.80	30.50
Qcells	Full	12 h	Depthwise TCN	13.07	40.05	67.40
Qcells	Power-active	1 h	Inverted-variate	7.56	11.29	33.00
Qcells	Power-active	12 h	Depthwise TCN	19.09	60.68	68.50

TABLE V. External prespecified Inverted-variate results: three-seed mean $\pm$ sample SD. nRMSE and skill are percentages; Daily skill uses its own common target intersection.

Site	Window	Scope	Train-range nRMSE	Last-value skill	Daily-matched skill
Yulara	1 h	full	9.36 ± 0.03	17.34 ± 0.26	36.28 ± 0.20
Yulara	1 h	power-active	12.72 ± 0.06	18.73 ± 0.36	37.28 ± 0.28
Yulara	4 h	full	12.16 ± 0.13	49.24 ± 0.53	17.25 ± 0.86
Yulara	4 h	power-active	16.06 ± 0.22	48.58 ± 0.70	20.83 ± 1.08
Yulara	8 h	full	14.28 ± 0.05	61.32 ± 0.14	2.71 ± 0.34
Yulara	8 h	power-active	19.03 ± 0.14	56.10 ± 0.31	6.10 ± 0.67
Yulara	12 h	full	14.37 ± 0.14	67.10 ± 0.32	1.62 ± 0.97
Yulara	12 h	power-active	19.28 ± 0.20	59.46 ± 0.42	4.50 ± 0.99
NIST Ground	1 h	full	6.91 ± 0.04	17.93 ± 0.52	57.73 ± 0.28
NIST Ground	1 h	power-active	11.01 ± 0.04	20.24 ± 0.26	59.01 ± 0.13
NIST Ground	4 h	full	11.05 ± 0.10	41.16 ± 0.51	32.09 ± 0.58
NIST Ground	4 h	power-active	17.22 ± 0.39	37.98 ± 1.39	36.07 ± 1.41
NIST Ground	8 h	full	15.59 ± 0.29	42.18 ± 1.06	3.88 ± 1.80
NIST Ground	8 h	power-active	21.58 ± 0.30	39.25 ± 0.85	20.22 ± 1.12
NIST Ground	12 h	full	17.29 ± 0.27	43.24 ± 0.87	-6.47 ± 1.65
NIST Ground	12 h	power-active	23.49 ± 0.40	37.80 ± 1.06	13.20 ± 1.49

minus mean Inverted is $+2.59$ kW with $[-0.62, 5.66]$ kW; this crossing interval does not establish equivalence. The original NIST full-target ordering therefore also changes. Full, active and low-power results, point counts and all fixed block lengths are retained separately for each period.

The NIST squared-error decomposition retains an active saving and a low-power cost, but their balance changes. The next-year mean seed contributions to Inverted-minus-Daily full MSE are -1275.75 and $+130.09$ kW², respectively, giving a net -1145.65 kW². These disjoint contributions explain the new full-target advantage without assigning low-power measurements to a particular weather or operating event.

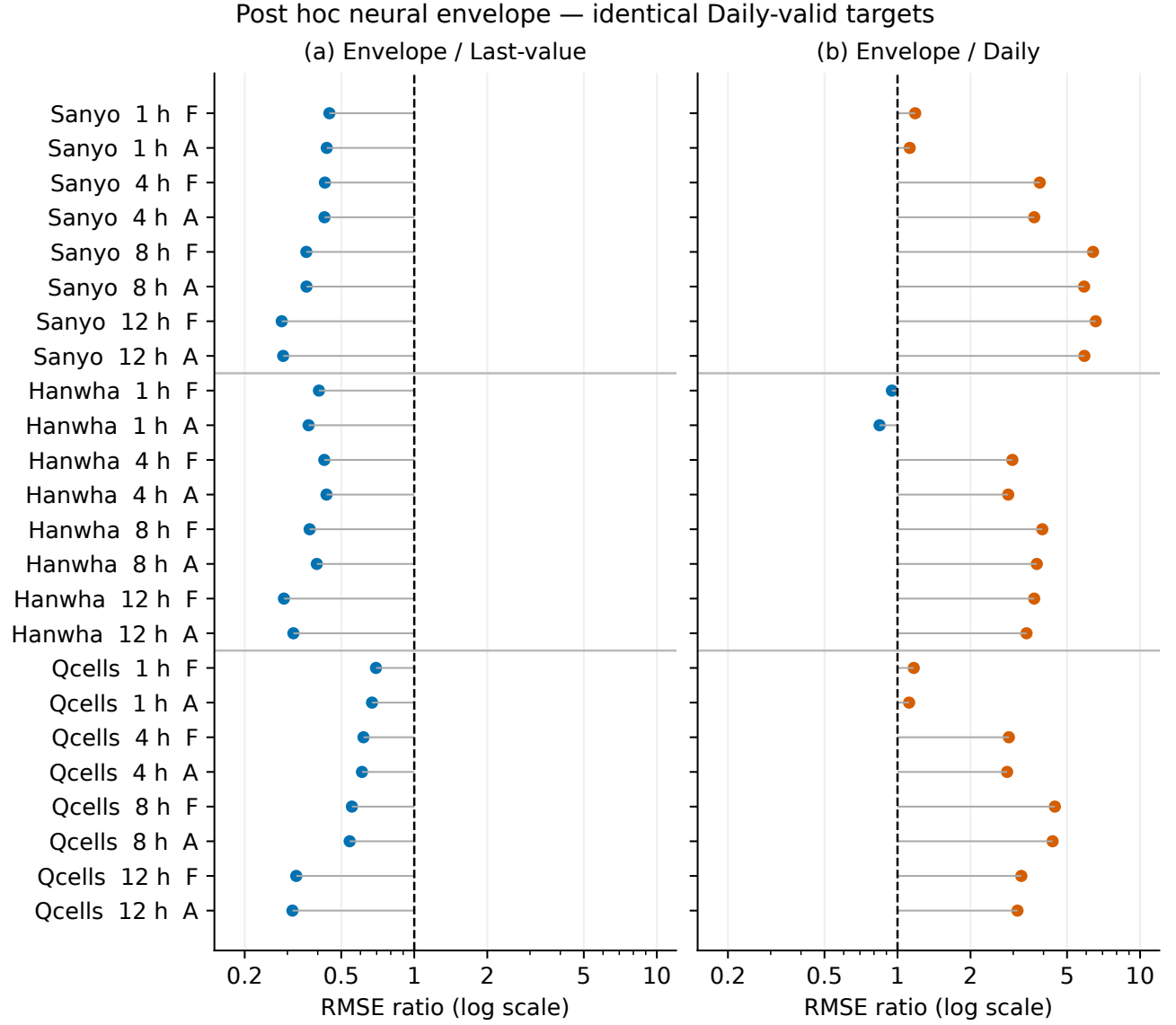


FIG. 5. Post hoc minimum three-seed mean neural RMSE relative to both references on the same Daily-valid targets. F: full; A: power-active. The envelope is descriptive, not deployable.

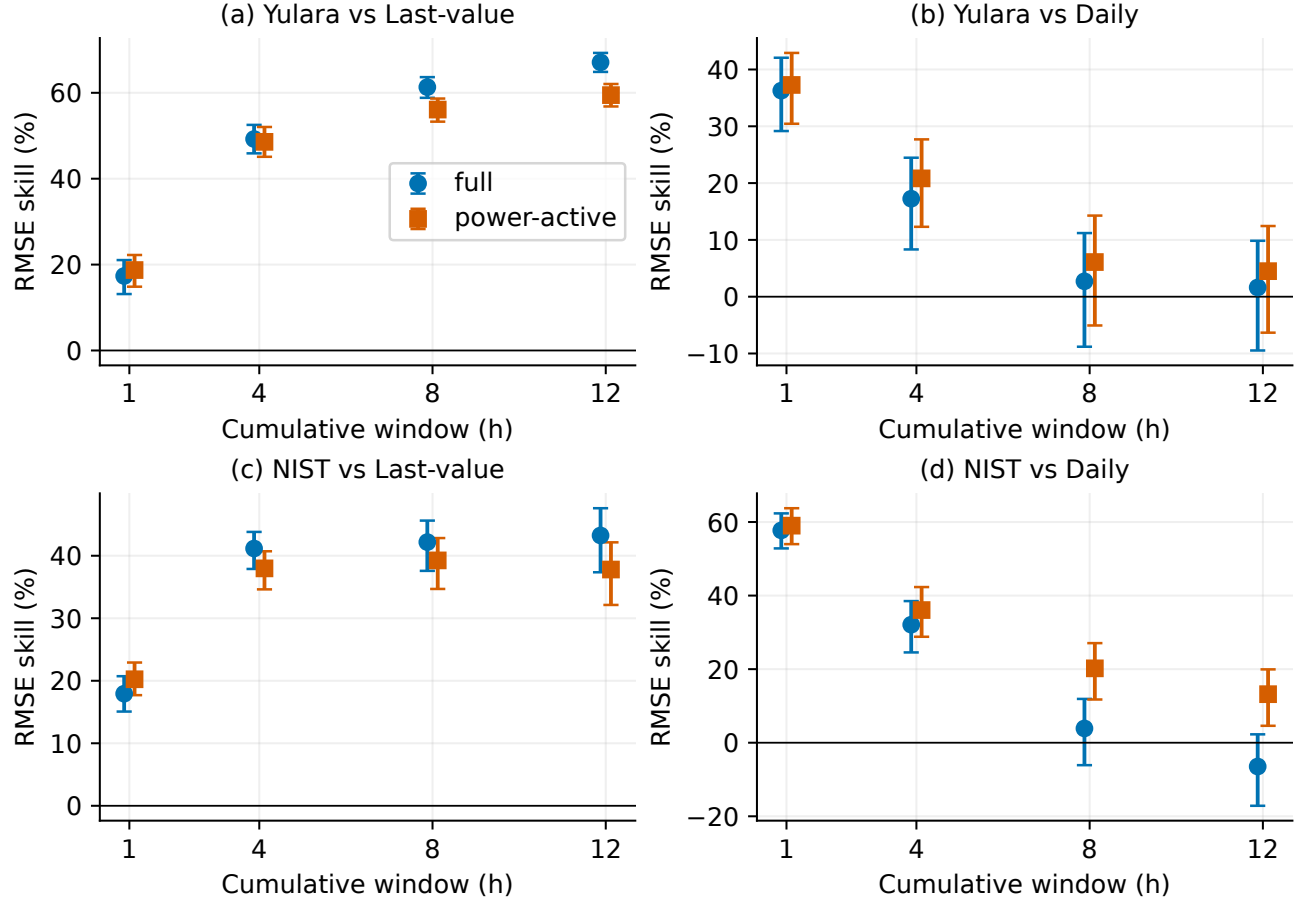


FIG. 6. Inverted-variate mean seed skill with 95% paired temporal-block intervals: 48 h blocks, 2000 resamples. These are conditional time-sample intervals, not seed SD. Each reference has its own matched targets.

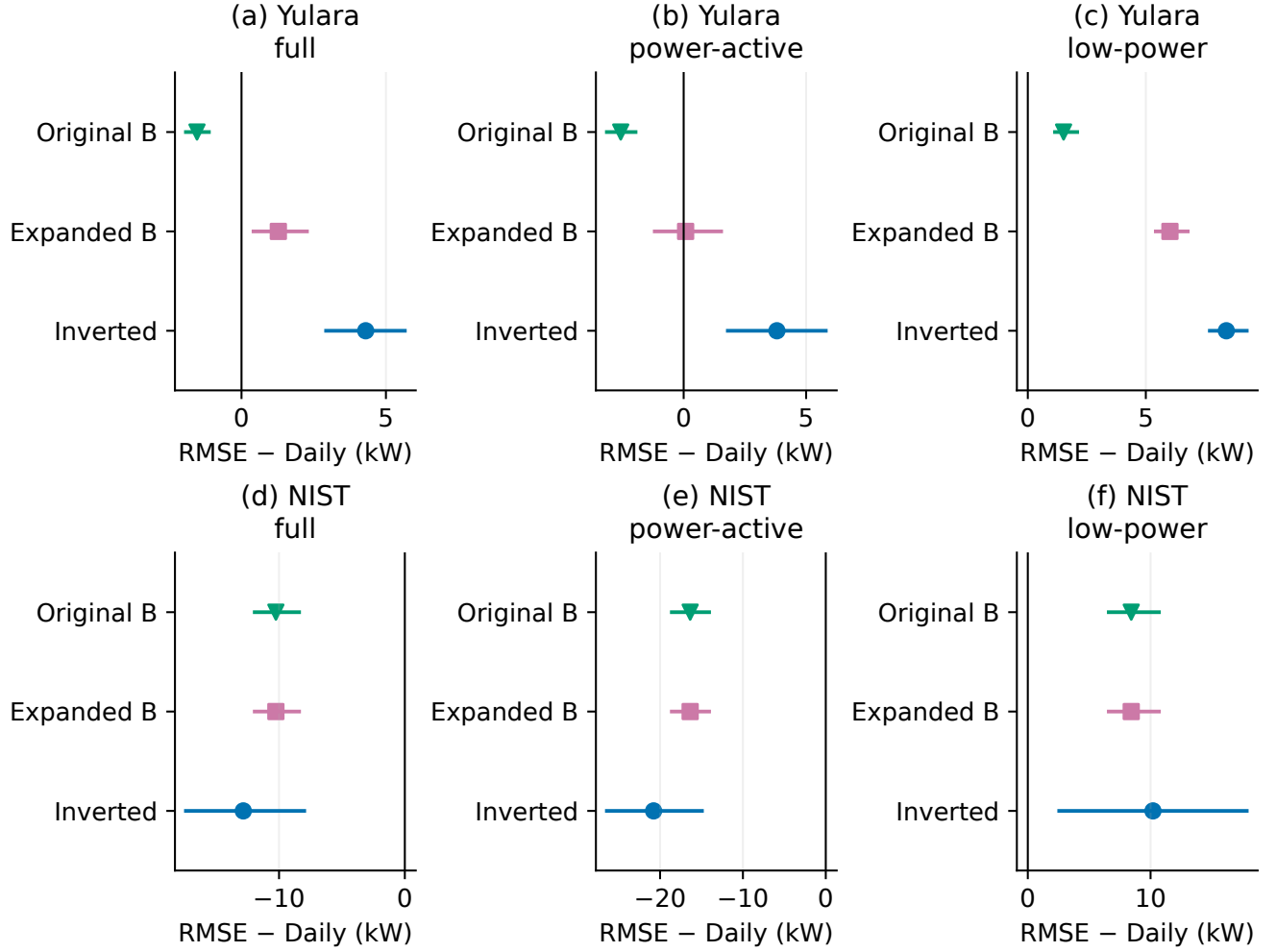


FIG. 7. Same-site next-year evaluation of cumulative twelve-hour windows, with origins fixed to April–June 2018. Negative differences favor the named method over Daily. Points and 95% paired 48-hour-block intervals use common finite target and reference support within each panel; Inverted averages seed-specific effects. Both saved Ridge grids are retained. Panel scales differ.

VI. INTERPRETATION AND LIMITS

Last-value holds the present power through the entire future trajectory, whereas learned recent-history models can represent changing output. On the original horizon-specific support, the repeated neural improvement shows added predictive value over that local-continuity reference; complete-window conditioning can change the population and the comparison. Daily supplies a measured trajectory from another day and therefore asks whether a model adds to daily recurrence. The Alice-external contrast and the daily-augmented Ridge results show why these questions should be reported separately. Differences in daily stability, operating distributions, and input availability are possible explanations for site differences, not mechanisms established by this experiment.

The power decomposition provides a direct explanation for one scope-dependent result: NIST’s active error saving is outweighed by low-power error. It also explains why the full-target winner need not be the active-target winner. This conclusion follows from disjoint squared-error contributions and does not require attributing the low-power records to night, snow, or downtime. Ranking differences across sites remain observational, with different years, input widths, target systems, and held-out months.

The three geographic sites broaden the evidence beyond a single facility. The original October–December performance splits and the added April–June next-year origin period do not constitute all-season validation. Period differences include model age, input exposure, operating conditions and scoring support, not an isolated seasonal effect. Alice’s three arrays are co-located and use 2018 data with 17 channels, compared with seven external channels in 2017. Each model was fitted separately; no new-period refitting is performed. The incomplete NIST operation logs and partial Yulara metadata limit event and sensor interpretation. Test records underwent data-quality inspection during development; they remain held-out performance splits, while the added decompositions, intervals, and Ridge comparisons are explicitly post hoc. Fixed-budget compact models do not represent fully tuned official implementations. Seed SD and finite-period block intervals address different, incomplete sources of uncertainty. Common target masks remove within-comparison sample discrepancies, not information-set differences or selection effects. The Qcells one-hour common-origin reversal demonstrates why the retained target distribution must also be reported.

VII. CONCLUSION

The benchmark identifies three conditions governing the interpretation of PV forecasting gains: reference history, target support and power composition, and the period in which frozen models are assessed. Daily

changes the conclusion drawn from Last-value at Alice, while matched squared-error decomposition explains the changing NIST balance between active savings and low-power costs. Added daily inputs and expanded regularization have distinct, period-dependent effects within Ridge. The fixed next-year results show that neither a neural model nor a selected Ridge grid has a uniformly favorable ordering. Qcells support counts further show how whole-window rejection can select a different target population. Reporting these conditions alongside paired temporal effects makes the evidence useful without treating an architecture name as a universal performance guarantee.

SUPPLEMENTARY MATERIAL

The Supplement adds common-origin and fixed-lead analyses, paired temporal-block intervals, target-power decompositions, deterministic Ridge references, monthly summaries and transparent trajectory cases alongside complete Alice evidence, external field and time audits, all 96 support groups, the 24-run matrix and checkpoint records, per-seed RMSE, complete mean/sample-SD errors, metadata limits, model configurations, and the descriptive envelope. Full per-seed metric CSVs accompany the code rather than being expanded into thousands of PDF rows.

ACKNOWLEDGMENTS

The authors acknowledge the DKA Solar Centre and NIST for providing the source photovoltaic and meteorological data.

FUNDING

Author confirmation required. Draft: This work is supported by the National Natural Science Foundation of China (Grant Nos. 62271151 and W2421092).

CONFLICT OF INTEREST

Author confirmation required. Draft: The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

ETHICS APPROVAL

Author confirmation required. Draft: This study analyzes publicly accessible photovoltaic and meteorological records and involves no human participants or animals; ethics approval was not applicable.

AUTHOR CONTRIBUTIONS

Author confirmation required. Draft: Jiangkun Zhu: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – Original Draft, Visualization. Mengling Yang: Validation, Writing – Review & Editing. Zhicong Chen: Conceptualization, Supervision, Project administration, Funding acquisition, Writing – Review & Editing. Lijun Wu: Supervision, Resources, Project administration, Funding acquisition, Writing – Review & Editing.

DECLARATION OF AI-ASSISTED TOOLS

Author confirmation required. The exact tool/version record and any journal-required disclosure wording remain pending author confirmation; no model designation is inferred.

DATA AVAILABILITY

Author confirmation required. Draft: Source observations are available from the official DKA Solar Centre services for Alice Springs and Yulara and from the NIST PV data service.^{18,20,23} Reproduction requires downloading data from those providers and complying with their terms. The code repository does not redistribute these raw observations. The final public code location and release details remain subject to author confirmation.

CODE AVAILABILITY

Author confirmation required. Draft: A dedicated public release of evaluation code, aggregate metrics, figure generation, and ordinary protocol tests is planned after manuscript approval and license selection. Raw data, checkpoints, and prediction arrays will not be redistributed. The final repository URL must be confirmed before submission.

- ¹J. Antónanzas, N. Osorio, R. Escobar, R. Urraca, F. J. Martínez-de Pison, and F. Antonanzas-Torres, “Review of photovoltaic power forecasting,” *Solar Energy* **136**, 78–111 (2016).
- ²A. Mellit, A. M. Pavan, and V. Lughi, “Deep learning neural networks for short-term photovoltaic power forecasting,” *Renewable Energy* **172**, 276–288 (2021).
- ³R. J. Hyndman and A. B. Koehler, “Another look at measures of forecast accuracy,” *International Journal of Forecasting* **22**, 679–688 (2006).
- ⁴P. Di Leo, A. Ciocia, G. Malgaroli, and F. Spertino, “Advancements and challenges in photovoltaic power forecasting: A comprehensive review,” *Energies* **18**, 2108 (2025).

- ⁵D. Markovics, G. B. Békési, and M. J. Mayer, “A unified benchmark of deep learning and classical machine learning models for weather-based solar power forecasting: Balancing complexity and skill,” *Energy and AI* **25**, 100833 (2026).
- ⁶J. Depoortere, J. Driesen, J. A. K. Suykens, and H. S. Kazmi, “SolNet: Open-source deep learning models for photovoltaic power forecasting across the globe,” *International Journal of Forecasting* **41**, 1223–1236 (2025).
- ⁷K. Mahmud, S. Azam, A. Karim, S. M. Zobaed, B. Shanmugam, and D. Mathur, “Machine learning based PV power generation forecasting in alic springs,” *IEEE Access* **9**, 46117–46128 (2021).
- ⁸S. Huang, Q. Zhou, J. Shen, H. Zhou, and B. Yong, “Multistage spatio-temporal attention network based on NODE for short-term PV power forecasting,” *Energy* **290**, 130308 (2024).
- ⁹H. S. E. Mansour, A. S. Mohamed, H. M. H. Farh, A.-W. Ibrahim, A. M. Al-Shaalan, A. B. Kunya, and H. S. Elnashar, “High-resolution photovoltaic power forecasting using machine learning models under seasonal and stress conditions,” *Scientific Reports* **16**, 19029 (2026).
- ¹⁰H. Hewamalage, C. Bergmeir, and K. Bandara, “Recurrent neural networks for time series forecasting: Current status and future directions,” *International Journal of Forecasting* **37**, 388–427 (2021).
- ¹¹Y. Liu, T. Hu, H. Zhang, *et al.*, “iTransformer: Inverted transformers are effective for time series forecasting,” in *International Conference on Learning Representations* (2024).
- ¹²Y. Nie, N. H. Nguyen, P. Sinthong, and J. Kalagnanam, “A time series is worth 64 words: Long-term forecasting with transformers,” in *International Conference on Learning Representations* (2023).
- ¹³D. Luo and X. Wang, “ModernTCN: A modern pure convolution structure for general time series analysis,” in *International Conference on Learning Representations* (2024).
- ¹⁴L. J. Tashman, “Out-of-sample tests of forecasting accuracy: An analysis and review,” *International Journal of Forecasting* **16**, 437–450 (2000).
- ¹⁵C. Bergmeir and J. M. Benítez, “On the use of cross-validation for time series predictor evaluation,” *Information Sciences* **191**, 192–213 (2012).
- ¹⁶H. Hewamalage, K. Ackermann, and C. Bergmeir, “Forecast evaluation for data scientists: Common pitfalls and best practices,” *Data Mining and Knowledge Discovery* **37**, 788–832 (2023).
- ¹⁷S. Kapoor and A. Narayanan, “Leakage and the reproducibility crisis in machine-learning-based science,” *Patterns* **4**, 100804 (2023).
- ¹⁸Desert Knowledge Australia Solar Centre, “Data download: Dkasc, alic springs,” (2026), official data service and system metadata; accessed 28 August 2026.
- ¹⁹K. Wang, X. Qi, and H. Liu, “A comparison of day-ahead photovoltaic power forecasting models based on deep learning neural network,” *Applied Energy* **251**, 113315 (2019).
- ²⁰DKA Solar Centre, “Yulara: 106.6 kw, mono-si, roof, 2016, sails in the desert—combined,” (n.d.), accessed September 7, 2026.
- ²¹M. Boyd, “Performance data from the NIST photovoltaic arrays and weather station,” *Journal of Research of the National Institute of Standards and Technology* **122**, 40 (2017).
- ²²National Institute of Standards and Technology, “NIST campus PV data dictionary v1.0,” (n.d.), accessed September 7, 2026.
- ²³National Institute of Standards and Technology, “NIST campus photovoltaic arrays and weather station data sets,” (n.d.), catalogue accessed September 7, 2026.
- ²⁴O. Troyanskaya, M. Cantor, G. Sherlock, *et al.*, “Missing value estimation methods for dna microarrays,” *Bioinformatics* **17**, 520–525 (2001).
- ²⁵F. T. Liu, K. M. Ting, and Z.-H. Zhou, “Isolation forest,” in *2008 Eighth IEEE International Conference on Data Mining* (2008) pp. 413–422.